Ganesh Chandrasekar

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10/03/2001   https://www.linkedin.com/in/cbsag/   https://github.com/cbsag

EDUCATION

Master of Computer Science (Thesis) — 09/2023 – present
Concordia University — Montreal, Canada

Bachelor of Engineering — 2018 – 2022
Anna University (KCG College of Technology) — Chennai, India
B.E. Computer Science and Engineering - **CGPA 8.81/10** (**First Class with Distinction**)

PUBLICATIONS

Agentic and Non-Agentic Multi-Hop Systems for Medical Question Answering — 14/08/2025
G. Chandrasekar, H. G. Saisudha, and S. Bergler
Proceedings of the BioCreative IX Challenge and Workshop (BC9): Large Language Models for Clinical and Biomedical NLP at the International Joint Conference on Artificial Intelligence (IJCAI), Montreal, Canada, 2025. Published Aug. 14, 2025. doi: 10.5281/zenodo.16875905

Exploration of Performance of Dynamic Branch Predictors used in Mitigating Cost of Branching — 18/10/2022
G. Chandrasekar , A. Ambashankar, C. A. R, and S. S
Proceedings of the 2022 Third International Conference on Intelligent Computing Instrumentation and Control Technologies (ICICICT), 2022, pp. 574–579, doi: 10.1109/ICICICT54557.2022.9917915

PROFESSIONAL EXPERIENCE

Research Assistant — 09/2023 – present
CLaC Lab, Concordia University — Montreal, Canada

- Research Assistant at the **CLaC Lab**, focusing on **information retrieval**, **biomedical text analysis**, and **retrieval-augmented generation (RAG)** using **LLMs** and **PLMs**.
- Designed systems integrating **retrieval techniques** with **grounded generation**, contributing to **MedHopQA (BioCreative IX)** and **TREC BioGen Task B** shared tasks.
- Built annotation and preprocessing pipelines using **GATE (General Architecture for Text Engineering), University of Sheffield**, for structured text analysis.
- Previously collaborated with the **Journalism Department** to develop retrieval workflows and support corpus-based news research.

Teaching Assistant — 09/2024 – present
Concordia University — Montreal, Canada

- Teaching Assistant for **COMP 479/6791 – Information Retrieval and Web Search**, guiding students through coursework, assignments, and project development.
- Assist in grading, clarifying core IR concepts, and supporting students in practical implementations.

OrangeScape Technologies Pvt Ltd — 07/2022 – 08/2023
Solution Developer — Chennai, India

- Developed software features for the **Kissflow Procurement Cloud** department.
- Built and integrated backend scripts within a **Low-Code/No-Code paradigm**.
- Worked on **workflow automation** and performed **software testing** to ensure robust deployments.

Basik Marketing Pvt. Ltd — 05/2022 – 07/2022
Software Engineering Intern (Backend) — Chennai, India

- Collaborated with the platform development team for **The Esports Club**, a marketing firm supporting the Indian gaming and esports industry.
- Evaluated platform specifications to create efficient backend solutions optimizing performance.
- Developed key backend components to enhance platform functionality and reliability.

Solarillion Foundation 01/2021 – 01/2022

Undergraduate Research Assistant

- Developed a Two-Stage Machine Learning Engine for **Flight Delay Prediction**.
- Worked on **Citation Intent Classification** and supervised text analysis tasks.

PALS: An Initiative by Alumni Fraternity of IITs 01/2021 – 06/2021

Software Engineering Intern

- Developed a simulator for the **Synchro-Transmitter & Receiver Characteristics Experiment** in the Control & Instrumentation Lab.
- The experiment was built under the supervision of **NIT Karnataka** and complied with **Ministry of Education** guidelines.

PROJECTS

TREC BioGen Task B: Biomedical Evidence-Based Generation Systems | Dashboard 🔗 2025

- Developed **retrieval-augmented generation (RAG)** systems integrating **sparse (BM25, TF-IDF)** and **dense (MedCPT)** retrieval with **Qwen-based LLMs** using **vLLM** for scalable inference.
- Designed multi-stage pipelines featuring **context summarization**, **MMR-based selection**, **cross-encoder reranking**, and **prompt engineering** to improve evidence grounding.
- Generated **answers with citations linked to PubMed articles**, ensuring transparency and verifiable biomedical evidence synthesis.
- Results visualized on an interactive dashboard.

Evaluating BERT-Based Models for Negation Detection in Biomedical Texts 2024

- Fine-tuned BERT, BioBERT, and ClinicalBERT on the BIOSCOPE dataset for negation and speculation detection using BIO tagging.
- Found BERT to outperform domain-specific models across token, span, and CRF Suite metrics, highlighting the robustness of general-purpose embeddings.

Government ID Based Single Sign-On Authentication System | Patent 🔗 2021 – present

- Proposed a **Single Sign-On Authentication System** that uses **Government IDs** for secure user verification.
- Designed a **transparent QR-based process** to validate user identity.
- Patent filed under the German Patent and Trademark Office

Undergraduate Project: Dynamic Branch Predictor 2022

- Part of a team of three developing a **deep-learning-based dynamic branch prediction engine** to improve **pipeline execution speed**.
- Conducted under the supervision of **Dr. Shounak Chakraborty**, **Marie Curie Post-Doctoral Fellow**, **NTNU**.

Flight Delay Prediction System 2021

- Built a **predictive ML engine** to classify delayed flights and forecast delay durations.
- Enabled airlines to take **proactive measures**, minimizing disruptions and improving **efficiency and customer satisfaction**.
- Performed **data cleaning**, **pre-processing**, and **model selection** for optimal results.

PROGRAMMING LANGUAGES AND SKILLS

Programming Languages

- Python, JavaScript, Java.

Libraries and Frameworks:

- PyTorch, Transformers, vLLM, Pandas, scikit-learn, Flask, React JS, Node JS, Nest JS, PWAs, SQL

Tools and Technologies:

- Git, GitHub, VS Code, Jupyter Notebooks, LaTeX, Figma, GATE (General Architecture for Text Engineering, University of Sheffield), Slurm (HPC), Conda
- Expertise in **retrieval-augmented generation**, **prompting**, **cross-encoder reranking**, **context selection**, and **pipeline orchestration** for **biomedical NLP**